14-Day Data Engineering Job-Focused Plan

Which phases of your uploaded Data Engineering Roadmap to study first

The short answer

Priority: Phase 1 + selected topics from Phase 2. **Then:** selected basics from Phase 3 and Phase 5. **Defer for now:** deep Phase 4, advanced Phase 5, and most Phase 6 projects. The original PDF estimates several months for the full roadmap, so this is a focused 14-day selection—not the entire roadmap.

Phase-by-phase decision

Roadmap phase	Decision	What to cover
Phase 1 — Foundations PDF pages 3–4	**Must cover — highest priority**	Python basics for data; CSV/JSON and file I/O; pandas basics; SQL SELECT/JOIN/GROUP BY/HAVING, CTEs and window functions; PostgreSQL; Linux command line; Git/GitHub.
Phase 2 — Data Pipelines & ETL PDF pages 5–6	**Must cover — core subset**	ETL vs ELT; files/APIs; cleaning and validation; loading into PostgreSQL; safe re-runs/upserts; basic Airflow concepts; dbt models/tests overview; retries and data quality basics.
Phase 3 — Data Warehousing & Storage PDF pages 7–8	**Selected basics only**	Fact vs dimension tables; star schema and grain; data warehouse vs data lake; Parquet concept; Bronze/Silver/Gold idea. Pick one warehouse platform for introductory familiarity.
Phase 4 — Distributed Processing PDF pages 9–10	**Defer deep study**	Know what Spark/PySpark and Kafka are and what problems they solve. Skip building streaming or large-scale distributed systems during this sprint unless a target job specifically requires them.
Phase 5 — Cloud & DataOps PDF pages 11–12	**Selected basics only**	Docker/Compose basics; cloud storage and IAM concepts; one cloud platform overview (e.g. AWS S3); environment variables; basic logging and failure handling. Defer Terraform, Kubernetes and advanced monitoring.
Phase 6 — Capstone Projects PDF pages 13–14	**Build one small project**	Build one end-to-end batch ETL pipeline and publish it on GitHub with README, setup steps, architecture diagram and data quality checks. Leave advanced lakehouse/streaming projects for later.

Suggested 14-day order

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1–2	Phase 1: Python for data, files, pandas	Read and clean a CSV/JSON dataset
3–5	Phase 1: SQL + PostgreSQL	Queries with joins, aggregation, CTEs and window functions
6–7	Phase 2: ETL with Python + PostgreSQL	Batch pipeline with validation and duplicate-safe loading
8	Phase 1: Linux + Git/GitHub	Run from command line; push project with README
9	Phase 3: data modelling basics	Simple e-commerce star schema
10	Phase 2: dbt fundamentals	Basic transformation models and tests, if time permits
11	Phase 2: Airflow fundamentals	Understand DAGs/tasks/schedules; create simple DAG if possible
12	Phase 5: Docker + cloud concepts	Understand packaging and deployment basics
13	Phase 6: polish one portfolio project	README, architecture diagram, validation and error handling
14	Interview review + applications	Explain project, revise SQL/Python, apply to jobs

What to postpone

For this sprint, do not try to master advanced Spark tuning, Kafka streaming, Delta Lake/Iceberg, complex Terraform stacks, Kubernetes, advanced observability dashboards, or multiple capstone projects. Return to these later or prioritize them if a specific job description requires them.

Daily study rule

Aim for about 4 hours daily if possible: 90 minutes learning, 2 hours hands-on work, and 30 minutes explaining the topic in your own words. If you have less time, keep the same priority order and reduce optional topics.

Basis: your uploaded Data Engineering Roadmap 2025 PDF—Phase 1 (pages 3–4), Phase 2 (pages 5–6), Phase 3 (pages 7–8), Phase 5 (pages 11–12), and Phase 6 (pages 13–14). This is a short job-preparation selection, not a replacement for the full roadmap.